React – Fundamentals

Session – 8

1. Create a React component called SearchBar with an input field and a button. Use useRef to focus the input field automatically when the component mounts.

Ans.

SearchBar:

```
import { useEffect, useRef } from "react";

function SearchBar() {
  const inputRef = useRef(null);

  useEffect(() => {
    inputRef.current.focus();
  }, []);

  return (
    <div>
      <input
        type="text"
        ref={inputRef}
        placeholder="Search..."
      />
      <button>Search</button>
    </div>
  );
}

export default SearchBar;
```

2. Build a simple login form with two controlled input fields (username and password) using useState. After the user clicks the 'Login' button, use useRef to clear and focus the username field.

Ans.

LoginForm.jsx:

```
import { useState, useRef } from "react";

function LoginForm() {
  const [username, setUsername] = useState("");
  const [password, setPassword] = useState("");
  const usernameRef = useRef(null);
```

```
  const handleLogin = () => {
    alert("Login Successful!");

    setUsername("");
    setPassword("");

    usernameRef.current.focus();
  };

  return (
    <div>
      <input
        type="text"
        ref={usernameRef}
        placeholder="Username"
        value={username}
        onChange={(e) => setUsername(e.target.value)}
      />

      <input
        type="password"
        placeholder="Password"
        value={password}
        onChange={(e) => setPassword(e.target.value)}
      />

      <button onClick={handleLogin}>Login</button>
    </div>
  );
}

export default LoginForm;
```

3. Make a React component called AddToPlaylist where users can enter a song name and click 'Add'. After adding, use useRef to focus the input again so users can quickly add another song.

Hint: Call inputRef.current.focus() after updating the playlist.

Ans.

AddToPlaylist:

```
import { useState, useRef } from "react";

function AddToPlaylist() {
  const [song, setSong] = useState("");
  const [playlist, setPlaylist] = useState([]);
  const inputRef = useRef(null);
```

```
  const handleAdd = () => {
    if (song.trim() === "") return;

    setPlaylist([...playlist, song]);
    setSong("");

    inputRef.current.focus();
  };

  return (
    <div>
      <input
        type="text"
        ref={inputRef}
        placeholder="Enter song name"
        value={song}
        onChange={(e) => setSong(e.target.value)}
      />

      <button onClick={handleAdd}>Add</button>

      <ul>
        {playlist.map((item, index) => (
          <li key={index}>{item}</li>
        ))}
      </ul>
    </div>
  );
}

export default AddToPlaylist;
```

4. Refactor an existing React form (for example, a feedback form with name and message fields) to use controlled components for both inputs, and add a button that, when clicked, focuses the message input using useRef.

Ans.

FeedbackForm:

```
import { useState, useRef } from "react";

function FeedbackForm() {
  const [name, setName] = useState("");
  const [message, setMessage] = useState("");

  const messageRef = useRef(null);
```

```
  const focusMessage = () => {
    messageRef.current.focus();
  };

  return (
    <div>
      <input
        type="text"
        placeholder="Enter your name"
        value={name}
        onChange={(e) => setName(e.target.value)}
      />

      <textarea
        ref={messageRef}
        placeholder="Enter your message"
        value={message}
        onChange={(e) => setMessage(e.target.value)}
      />

      <button onClick={focusMessage}>Focus Message</button>
    </div>
  );
}

export default FeedbackForm;
```